

Disproof of conjecture 00000009984: squared irreducible dimensions need not divide $\dim H$

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1 The conjecture

English. Definition: Semisimple Hopf algebras: finite-dimensional semisimple coalgebra structures.
Conjecture: The sum of squared dimensions of irreducible modules of a semisimple Hopf algebra factors as the unique factorization of the dimension, and each squared factor divides the dimension; the strengthened uniqueness of the factorization excludes alternating-type exceptions, and the exception list is empty. (Hopf module dimension square factorization)

Chinese text (as in the repository): the sum of the squares of the dimensions of the irreducible modules decomposes as the unique decomposition of the dimension, and *each square factor divides the dimension*; the exception list is empty.

2 Reading

The conjecture is a conjunction. We refute one conjunct, the *divisibility clause*:

(D) For every finite-dimensional semisimple Hopf algebra A over $\mathbb{C}$ and every irreducible (simple) left A -module M , the square $(\dim_{\mathbb{C}} M)^2$ divides $\dim_{\mathbb{C}} A$.

Here “squared factor” is read as the square d_i^2 of an irreducible dimension d_i , i.e. a summand of the “sum of squared dimensions” $\sum_i d_i^2$. “The exception list is empty” means (D) is asserted for every such A , with no exceptions. Since the conjunction contains (D), a counterexample to (D) refutes the conjecture as read. The first conjunct ($\sum_i d_i^2 = \dim A$) is not used.

Not refuted: the different reading “each irreducible dimension d_i (unsquared) divides $\dim A$ ”. Our example satisfies that reading ($d_i \in \{1, 2\}$, $\dim A = 6$), and this submission makes no claim about it.

3 Definitions

- A *Hopf algebra* over $\mathbb{C}$ is a bialgebra $(A, m, u, \Delta, \varepsilon)$ with an antipode S satisfying $m(S \otimes \text{id})\Delta = u\varepsilon = m(\text{id} \otimes S)\Delta$. In Lean this is Mathlib’s class `HopfAlgebra` $\setminus \mathbb{C} A$.
- A is *semisimple* if it is semisimple as an algebra, i.e. every left A -module is a direct sum of simple modules (Lean: `IsSemisimpleRing A`).
- An *irreducible* module is a simple left A -module: nonzero with no submodules other than 0 and itself (Lean: `IsSimpleModule A M`). Its $\mathbb{C}$ -structure is the restriction of the A -structure (Lean: `IsScalarTower` $\setminus \mathbb{C} A M$); we also require it to be finite-dimensional, which only weakens (D) and so strengthens the refutation.

4 The counterexample $A = \mathbb{C}[S_3]$

Let S_3 be the symmetric group on $\{0, 1, 2\}$ and $A = \mathbb{C}[S_3]$ its group algebra, with the standard Hopf structure $\Delta(g) = g \otimes g$, $\varepsilon(g) = 1$, $S(g) = g^{-1}$ (Mathlib's instance on `MonoidAlgebra`). Then:

1. $\dim_{\mathbb{C}} A = |S_3| = 6$ (Lean: `finrank_H`).
2. A is semisimple by Maschke's theorem, since $6 \neq 0$ in $\mathbb{C}$ (Lean: `H_semisimple`, from Mathlib's Maschke instance).
3. Every $g \in S_3$ is group-like: $\Delta(g) = g \otimes g$, $\varepsilon(g) = 1$, $S(g) = g^{-1}$ (Lean: `grouplike`). Hence the coalgebra A is the direct sum of the one-dimensional subcoalgebras $\mathbb{C}g$, so A also has a semisimple coalgebra structure, as in the conjecture's definition line.

S_3 is a symmetric group, so A is not of “alternating type” under any meaning of that phrase; in any case the conjecture says the exception list is empty.

Lemma 1 (`sq_dvd_six`). *If $d \in \mathbb{N}$ and $d^2 \mid 6$ then $d = 1$.*

Proof. $d^2 \leq 6$ gives $d \leq 2$; $d = 0$ fails since $0 \nmid 6$, and $d = 2$ fails since $4 \nmid 6$. □

Lemma 2 (`comm_on_simple`). *Let $N \subseteq A$ be a simple left ideal with $\dim_{\mathbb{C}} N = 1$. Then $(ab - ba)n = 0$ for all $a, b \in A$ and $n \in N$.*

Proof. Pick $0 \neq v \in N$; then $N = \mathbb{C}v$, so $av = \alpha v$ and $bv = \beta v$ for scalars α, β . Since scalars are central in A , $abv = \alpha\beta v = bav$, so $(ab - ba)v = 0$, and hence $(ab - ba)(cv) = c(ab - ba)v = 0$. □

Theorem 3 (`H_not_squaredDimsDivide`). *$A = \mathbb{C}[S_3]$ violates (D).*

Proof. Assume (D) for A . Every simple left ideal N of A is a simple A -module, so $(\dim N)^2 \mid 6$ and $\dim N = 1$ by the first lemma. By the second lemma, every commutator $c = ab - ba$ annihilates every simple left ideal. Since A is a semisimple module over itself, it is the supremum of its simple submodules (Mathlib: `IsSemisimpleModule.sSup_simple_eq_top`), and the annihilator of a supremum is the intersection of the annihilators (`Submodule.annihilator_iSup`). So c annihilates A , and $c = c \cdot 1 = 0$. Thus A is commutative. But for the transpositions $\sigma = (01)$, $\tau = (12)$ we have $\sigma\tau \neq \tau\sigma$, and distinct group elements are distinct basis vectors of A , so $\sigma\tau \neq \tau\sigma$ in A (Lean: `H_not_comm`). Contradiction. □

(Concretely, the offending simple module is the 2-dimensional standard representation; the Lean proof does not need to construct it.)

Theorem 4 (`disproof`). *It is not true that every finite-dimensional semisimple Hopf algebra A over $\mathbb{C}$ satisfies (D).*

5 Lean formalization

File `lean/Conjecture9984/Basic.lean` (Lean 4, Mathlib v4.33.1, 127 lines). Definition:

```
def SquaredDimsDivide (A : Type) [Ring A] [HopfAlgebra C A] : Prop :=
  forall (M : Type) [AddCommGroup M] [Module A M] [Module C M]
    [IsScalarTower C A M] [IsSimpleModule A M] [FiniteDimensional C M],
    (finrank C M) ^ 2 | finrank C A
```

Main theorem:

```
theorem disproof :  
  not (forall (A : Type) [Ring A] [HopfAlgebra C A] [IsSemisimpleRing A]  
    [FiniteDimensional C A], SquaredDimsDivide A)
```

(here C is $\mathbb{C}$). Supporting theorems: `finrank_H`, `H_semisimple`, `grouplike`, `H_not_comm`, `sq_dvd_six`, `comm_on_simple`, `H_not_squaredDimsDivide`.

Axiom audit. `#print axioms C9984.disproof` (likewise `grouplike`, `finrank_H`) reports only `propext`, `Classical.choice`, `Quot.sound`. There is no `sorry` and no `native_decide`. `lake build` succeeds.